THE MINIMUM NUMBER OF COPRIME ADJACENCIES IN A PERMUTATION

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Abstract

We determine exactly the minimum number of coprime adjacent pairs in a permutation of $\{1, \dots, n\}$. The proof gives an optimal path cover of the graph joining integers with a common prime factor. In particular, the minimum is asymptotic to $7n/(12 \log n)$.

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The arithmetic properties of consecutive entries of permutations were studied by Erdős, Freud and Hegyvári [1]. Related problems involving coprime permutations and path partitions of arithmetic graphs have subsequently been investigated in [2, 3, 4, 5]. Here we consider the minimum number of coprime adjacent pairs in a permutation of $\{1, \dots, n\}$.

This problem is naturally expressed through the graph whose vertices are the integers $1, \dots, n$, with two distinct vertices adjacent when they have a common prime factor. The number of connected components of this graph is known [6]. We determine its minimum path-cover number exactly.

For a permutation $a = (a_1, \dots, a_n)$, put

$$h(n) = \min_{a \in S_n} \#\{1 \le i < n : \gcd(a_i, a_{i+1}) = 1\}.$$

Write $\pi(x)$ for the number of primes not exceeding x.

Theorem. *For every integer $n \ge 4$,*

$$\boxed{h(n) = \pi(n) - \pi(n/2) + \max\left\{1, \left\lceil \frac{\pi(n/2) - \pi(n/3)}{2} \right\rceil\right\}}. \tag{1}$$

Moreover, $h(1) = 0$, $h(2) = 1$, $h(3) = 2$, and

$$h(n) \sim \frac{7n}{12 \log n}. \tag{2}$$

Proof. Let H_n be the graph on $\{1, \dots, n\}$ in which distinct vertices u, v are adjacent precisely when $\gcd(u, v) > 1$. Denote by $\mathrm{pc}(G)$ the minimum number of vertex-disjoint paths covering a graph G, allowing singleton paths.

Cutting a permutation at its coprime adjacencies, and conversely concatenating the paths of a minimum cover, gives

$$h(n) = \mathrm{pc}(H_n) - 1. \tag{3}$$

Put

$$A = \pi(n) - \pi(n/2), \qquad B = \pi(n/2) - \pi(n/3).$$

The vertex 1 and the A primes in $(n/2, n]$ are isolated. Let K_n be the induced graph obtained by deleting them. Since K_n contains the adjacent vertices $2, 4$ for $n \ge 4$, at least one path is necessary to cover it.

Moreover, each prime $q \in (n/3, n/2]$ has exactly one neighbor in K_n, namely $2q$. These B degree-one vertices must occur at the ends of paths, with at most two in any one path. Consequently,

$$\mathrm{pc}(K_n) \ge r := \max\{1, \lceil B/2 \rceil\}. \tag{4}$$

We show that equality holds.

Suppose first that $n \ge 12$. For each prime $p \le n/2$, let

$$C_p = \{t \in \{2, \dots, n\} : P^+(t) = p\},$$

where $P^+(t)$ is the largest prime factor of t. These classes partition the vertices of K_n, and every C_p induces a clique.

For $p \le n/4$, the vertices $2p$ and $4p$ belong to C_p. We may therefore arrange every such class as a path with two even endpoints.

For $n/4 < p \le n/3$, we have $p \ge 5$ and

$$C_p = \{p, 2p, 3p\}.$$

Arrange each such class as a path from $2p$ to $3p$. Pair these paths arbitrarily, joining their endpoints $3p$ and $3q$, which have common factor 3. Each pair produces a path with two even endpoints.

If the number of these intermediate classes is odd, join the unpaired class C_p to C_3 instead. Since $n \ge 12$, the class C_3 contains both 6 and 9. Arrange C_3 from 6 to 9, then join 9 to $3p$ and traverse C_p toward $2p$. This again produces a path with two even endpoints, using C_3 exactly once.

Writing $s = \pi(n/4)$ and $m = \pi(n/3) - \pi(n/4)$, we have thus covered all classes with $p \le n/3$ by exactly $u = s + \lfloor m/2 \rfloor$ vertex-disjoint paths, each with two even endpoints. In the odd case, the combined path replaces the individual C_3-path.

Any two such paths may be joined through their even endpoints, producing another path with two even endpoints. It therefore suffices to show that $u \ge r$.

The classical prime-counting inequality [7, 8]

$$\pi(2x) \le 2\pi(x) \qquad (x \ge 3)$$

gives, since $n \ge 12$,

$$B \le \pi(n/2) \le 2\pi(n/4) = 2s.$$

As $s \ge 1$, we obtain $r \le s \le u$.

We may consequently merge the u paths into exactly r paths, each having two even endpoints.

Finally, for every prime $q \in (n/3, n/2]$,

$$C_q = \{q, 2q\}.$$

Attach these B classes to the $2r$ available even endpoints, at most one class per endpoint, using the edge from that endpoint to $2q$. Such an edge exists because both vertices are even. Since $B \le 2r$, all classes can be attached without creating additional paths.

We have constructed a cover of K_n by exactly r paths. Together with (4), this proves

$$\mathrm{pc}(K_n) = r. \tag{5}$$

Adding the $A + 1$ isolated vertices gives

$$\mathrm{pc}(H_n) = A + 1 + r,$$

which proves (1) for $n \ge 12$.

For $4 \le n \le 11$, equality is attained by the following permutations:

n	Permutation	$h(n)$
4	1, 2, 4, 3	2
5	1, 2, 4, 3, 5	3
6	1, 2, 4, 6, 3, 5	2
7	1, 2, 4, 6, 3, 5, 7	3
8	1, 2, 4, 8, 6, 3, 5, 7	3
9	1, 2, 4, 8, 6, 3, 9, 5, 7	3
10	1, 3, 9, 6, 2, 4, 8, 10, 5, 7	2
11	1, 3, 9, 6, 2, 4, 8, 10, 5, 7, 11	3

The cases $n \le 3$ are immediate.

Finally, the prime number theorem gives

$$A \sim \frac{n}{2\log n}, \qquad B \sim \frac{n}{6\log n},$$

and hence (2). □

Remarks. (1) The connected-component count of H_n is $A + 2$ for $n \ge 4$ [6]. Our theorem shows that

$$\mathrm{pc}(H_n) - (A + 2) = \max\{1, \lceil B/2 \rceil\} - 1.$$

Thus the known component count does not generally determine the minimum path cover.

(2) The function $h(n)$ is not monotone. For instance, $h(5) = 3$ while $h(6) = 2$.

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